

GEOMIND-R

Generative inverse design of geopolymers for the immobilisation of radioactive Cs-137 and Sr-90

Third progress report. The evidence for the project's central chemical finding is now complete on three independent levels — correlation, a within-sample causal test, and a direct atomic mechanism — the last supplied by the very paper requested at Checkpoint 2.

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Date	21 July 2026
Report status	Checkpoint 3 · literature-scale meta-analysis phase

Executive summary

The central finding now rests on three independent legs

Between Checkpoints 1 and 3 the project's core claim — that *framework aluminium*, not surface area, governs how a geopolymer captures Cs^+ and Sr^{2+} — has been raised from a plausible correlation to a result supported on three levels at once. Checkpoint 3 reports the third and strongest: a **direct atomic-scale mechanism**, delivered by the Geddes 2025 paper you supplied.

141 Pool A rows (14 sources)	54 Pool B rows (8 sources)	35 audited findings (28 resolved)	125 automated integrity tests passing
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THE EVIDENCE CHAIN, NOW COMPLETE

1. Correlation.

Across every dataset that separates them, framework aluminium tracks cation uptake (up to +0.95) while surface area does not (pooled +0.08).

2. Causation (within-sample).

Leaching a set of geopolymers strips framework Al, and uptake falls with it on the *same* samples: $\text{corr}(\Delta[\text{Al}^{\text{IV}}], \Delta K_{\text{D}}) = +0.938$.

3. Mechanism (atomic).

X-ray absorption spectroscopy and multinuclear NMR show Sr^{2+} sitting at the **charge-balancing sites of tetrahedral Al^{IV}** itself — the exact sites the descriptor counts.

WHAT CHANGED SINCE CHECKPOINT 2

1. The requested bridge paper arrived and closed the mechanism question.

Geddes 2025 — the priority-1 acquisition from the Checkpoint 2 request — images Sr binding at the Al^{IV} sites directly (Section 2).
2. Two more immobilisation sources were added,

one of them showing that a common accelerated leach test *hides* the caesium/strontium difference (Section 1).
3. The honest gap is now precisely defined.

The mechanism is settled; what remains scarce is *breadth* — composition-varying data pairing the descriptor with a target — and Section 5 states exactly what would close it.

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Section 1

What has advanced since Checkpoint 2

Three papers were received; the first is the one that mattered most.

Paper	What it added	Effect
Geddes 2025 (J. Hazard. Mater.) — the Checkpoint 2 priority request	the atomic mechanism : Sr^{2+} binds at Al^{IV} charge-balancing sites (EXAFS + NMR)	mechanistic capstone (Section 2)
Frederickx 2025 (Sustainability)	Cs/Sr leachability of a real waste immobilised in a metakaolin N-A-S-H form	Pool B +4 (leachant-flagged)
Kim 2026 (clean manuscript)	duplicate of a source already ingested	skipped

A METHODOLOGICAL POINT WORTH CARRYING FORWARD (FREDERICKX)

Frederickx's waste form passes the regulatory leach criterion, but in the *accelerated* test (an aggressive NH_4NO_3 medium) caesium and strontium score almost the same. That medium was designed for calcium-rich cements and preferentially attacks divalent cations, so it **understates** strontium immobilisation. The authors' own deionised-water extrapolation restores the usual, large Cs/Sr gap (leachability index 9.6 for Cs vs 12.5 for Sr). The lesson — recorded in our database as a leachant flag so these values are never mixed with deionised-water results — is that *the choice of leach test can invert the apparent ranking of a wasteform*. This matters for how any future comparison is framed.

Section 2

The central finding — a complete evidence chain

The project's central chemical result is that cation uptake in a geopolymer is *ion exchange at the charge-balancing sites created by framework aluminium*, not physisorption on a surface. It is now supported at three independent levels.

2.1 Level 1 — correlation (across samples)

Wherever the two can be separated, the density of framework aluminium tracks uptake strongly and positively, while BET surface area does not:

Varon (n=7, Sr)	corr(Al^{IV} , K_{D}) +0.95 vs corr(BET, K_{D}) +0.19
Oulu (n=4, NH_4^+)	corr(framework-Al index, uptake) +0.93
All cations pooled	corr(BET, uptake) +0.08 — surface area carries essentially no signal
Organic dyes (contrast)	corr(BET, uptake) +0.66 — opposite sign; a different, surface-binding mechanism

2.2 Level 2 — causation (within the same samples)

A companion study leached seven geopolymers with water, stripping framework aluminium, and re-measured strontium uptake on the identical samples. The samples that lost the most framework Al lost the most sorption:

corr($\Delta[\text{Al}^{\text{IV}}]$, ΔK_{D}) = **+0.938** (n = 7)

and the absolute density of exchange sites, $[\text{Al}^{\text{IV}}]$ in mmol g^{-1} , predicts K_{D} at **+0.946** (+0.986 excluding one barely-reacted outlier). This is a within-sample test, not merely a correlation across different materials — the strongest causal evidence a literature meta-analysis can provide.

2.3 Level 3 — mechanism (at the atomic scale) — the new result

The Geddes 2025 paper you supplied closes the chain. Using strontium K-edge X-ray absorption spectroscopy together with ^{29}Si , ^{27}Al and ^{23}Na NMR on metakaolin geopolymers, it shows that at low loading Sr^{2+} is incorporated into **extra-framework charge-balancing sites bonded through oxygen to tetrahedral aluminium** ($\text{Si}^{\text{IV}}\text{--O--Al}^{\text{IV}}$), with a local structure resembling the mineral brewsterite. In other words, strontium is observed sitting *exactly at the Al^{IV} sites the descriptor counts*. Above a chemical binding limit ($\approx 0.5\text{--}1.5$ wt.% $\text{Sr}(\text{OH})_2\cdot 8\text{H}_2\text{O}$), the excess precipitates as SrCO_3 rather than binding to the framework.

WHY THIS MATTERS

Correlation and even a causal test leave open the question of *why*. Geddes answers it directly: the framework-aluminium descriptor works because tetrahedral Al^{IV} is the strontium binding site, imaged at the atomic scale. The three levels — a statistical correlation, a within-sample perturbation experiment, and a spectroscopic mechanism — are mutually independent and point to the same conclusion. This is the mechanistic capstone of the descriptor-and-methodology paper.

2.4 The immobilisation side agrees, and the structural precondition holds

Independently of the adsorption data, caesium leach resistance stratifies by binder chemistry exactly as the framework picture predicts: metakaolin framework $\approx 13 >$ fly-ash framework $\approx 10 >$ calcium-rich slag ≈ 7.5 . And the descriptor remains meaningful only where a condensed aluminosilicate framework exists: in calcium-rich binders the silicon network collapses to low-connectivity environments and no compositional descriptor predicts uptake. The rule stands — never pool across structural classes.

Section 3

Current data structure

Two parallel datasets, never merged, both governed by a single machine-checked registry (17 assets, 35 findings, 16 decisions). Every row of both pools is listed in Appendices B and C.

POOL A — ADSORPTION (141 ROWS, 14 SOURCES)

Target: adsorption capacity or distribution coefficient; 75 rows are radionuclide (Cs/Sr).

Source	rows
oulu2026	63
literature_compilation	21
varon2025	7
varon_leached2026	7
lin2026	6
katada2024	6
tarnovsky2024	6
kurumisawa2021	5
xiang2021	5
petlitckaia2020	4
qian2001	4
baek2018	3

Source	rows
tian2019_espr	2
zhang2021	2

POOL B — IMMOBILISATION (54 ROWS, 8 SOURCES)

Target: leach resistance. Now spans metakaolin, fly-ash and slag binders, with an explicit leachant flag after Section 1's finding that leach-test chemistry can invert the apparent ranking.

Source	rows
jang2016	14
kurumisawa2021	10
nevin2026	10
arbelhaddad2022	6
kim2026	6
frederickx2025	4
komljenovic2020	2
stanojevic2025	2

Section 4

What we still cannot do, and why

The mechanism question is now answered. The remaining obstacle is different, and it is important to state it precisely so that the requests in Section 5 are the right ones.

4.1 Mechanism is settled; breadth is not

Geddes proved *why* the descriptor works, but on samples of a single fixed composition — it did not add data in which composition *varies* while both the descriptor and a Cs/Sr target are measured. That composition-varying paired data is what a generative model must learn from, and it remains scarce: essentially one strontium sample-family (seven compositions) plus one ammonium series (four). A model trained on so little, drawn from many different laboratories, would learn inter-laboratory differences and report them as chemistry. **Confidence in the mechanism has risen sharply; the breadth needed to train the generative extension has not.**

4.2 The faithful replication is still blocked on confidential data

The reference model was trained on 112 samples from one laboratory; only ten are public. A faithful replication needs the full set, which is confidential and must be requested.

4.3 Therefore the near-term output is a paper, not a model

With a correlation, a causal test and now an atomic mechanism, the descriptor-and-methodology result is strong and publishable now. The generative model remains gated on either the confidential dataset or a substantially broader body of composition-varying paired data — precisely what Section 5 asks for.

Section 5

Remaining acquisitions & data needs (the requests)

The Checkpoint 2 top request has been fulfilled — thank you. The remaining needs have narrowed to a single, well-defined gap: **breadth**.

5.1 Fulfilled since Checkpoint 2

Request	Status
Geddes 2025 — <i>Sr incorporation mechanism & binding capacity</i> (J. Hazard. Mater. 488, 137426)	✓ DELIVERED — supplied by the supervisor; closed the mechanism question (Section 2.3)

5.2 Still needed

Priority	What to obtain	Why it matters now	Unblocks
1 — the breadth gap	Composition-varying studies pairing an NMR framework-Al descriptor with a Cs/Sr capacity / K_d / leachability — the more distinct Si/Al compositions in one study, the better	The mechanism is settled; what a generative model still lacks is <i>range</i> . Named candidate: <i>Next Sustainability</i> 2025 (article S2950631X25000024), the one source with a 1–5 M activator axis carrying Cs, Sr and NMR together.	M4, M5
2 — faithful replication	The GEOMIND 112-sample training dataset — request to the reference-paper authors (RSC <i>Digital Discovery</i> , DOI 10.1039/d5dd00383k); a drafted letter already exists	The only route to a faithful replication of the reference model.	M3a

5.3 Data-types needed — and what counts as an acceptable substitute

If the named source cannot be obtained, an equivalent meeting these criteria gives the same insight — use them as search criteria.

Data-type we still need	Minimum acceptable form (a valid drop-in substitute)
Composition-varying descriptor + target — now the single core need	Any study giving a ^{29}Si Q^4 (mAl) deconvolution <i>or</i> a ^{27}Al Al ^{IV} fraction and a Cs or Sr capacity / K_d / leachability on the same ≥ 4 samples that differ in Si/Al . Metakaolin, fly-ash or slag are all acceptable if stated. (Geddes met the descriptor-plus-target criterion but at one fixed composition — what is missing is the <i>variation</i> .)
A second leaching / degradation series	Framework Al and uptake measured on the same samples before and after a treatment — an independent instance of the within-sample causal test would further harden the result.
A single-lab, many-sample formulation set	≥ 50 geopolymer formulations from one laboratory with measured properties (and ideally a Cs/Sr metric) — the minimum scale for training a generative model without inter-laboratory artefacts.

5.4 Where such data is most likely to be found

The two groups that publish almost all paired descriptor-and-target data remain the best hunting ground: the **Sheffield group** (Walkley, Provis, Geddes, Nevin, O'Donoghue, Stanojević, Komljenović — the source of the mechanism paper just delivered) and the **CEA Montpellier group** (Poulesquen, Varon, Gossard, Petlitckaia). Useful venues: J. Hazard. Mater. , J. Nucl. Mater. , Cement & Concrete Research , Dalton Transactions , Open Ceramics . Any of their studies that varies Si/Al across several samples while reporting both NMR and a Cs/Sr metric would directly close the breadth gap.

Section 6

Plan and requests

6.1 Ready to write up now

The descriptor-and-methodology paper is ready in substance: framework aluminium predicts cation uptake (correlation), losing it costs uptake on the same sample (causation), and strontium is imaged binding at the Al^{IV} sites (mechanism) — plus a saturation screen that exposes published capacities that are extrapolation artefacts. None of this needs the confidential dataset.

6.2 What we ask of this checkpoint

- **Help close the breadth gap** — obtain *Next Sustainability* 2025, or identify any study from the Section 5.4 groups that varies Si/Al while reporting NMR and a Cs/Sr metric.
- **Support the confidential-data request** to the reference-model authors.
- **Advise on a small own-laboratory measurement** — a short Si/Al series with NMR and a Sr uptake test would, by itself, supply the composition-varying paired data the model needs and let us replicate the causal test in-house.
- **Agreement** to proceed to writing the descriptor-and-methodology paper, with the generative model as the subsequent phase once breadth allows.

Appendix A

Full finding register

All 35 audited findings, most-recent last, each with the evidence that established it and the action taken. Severity is the data-integrity impact.

critical F1 — Si_Al and Density transplanted from the GEOMIND paper [resolved] EVIDENCE 9 of 11 unique Si_Al values match the paper's 10 samples exactly to 3 dp; all 10 published density values appear; 91 of 92 mixes carry a single broadcast value; neither column exists in either upstream source file. ACTION Both columns dropped in clean.py. Must never be reinstated as features.
critical F2 — Dataset far smaller than 867 rows implies [resolved] EVIDENCE Only 92 distinct mix designs; Metakaolin constant zero; features have 2-6 unique values. ACTION Grouped CV by mix design is mandatory. Metakaolin dropped.
high F3 — Adsorption data has extensive internal duplication [resolved] EVIDENCE 24 Qmax_Cs rows contain only 16 distinct values; Qmax_Sr has 7 rows. ACTION Repeats flagged (not deleted) via cs/sr_value_repeated.
medium F4 — Redundancy and version disagreement [partially resolved] EVIDENCE CSV/XLSX byte-identical; 18 duplicate rows; v2 disagrees in 22 cells. ACTION CSV declared canonical; duplicates removed; v2 unresolved.
low F5 — Metadata.csv is cp1256, not UTF-8 [resolved] ACTION Handled in the ingestion layer.
critical F6 — Feasibility_Ranges is mislabelled and anti-correlated with truth [resolved with open gap] EVIDENCE Rejects 7 of 9 feasible published samples, accepts the 1 infeasible. Its constants are verbatim from the retired tool at tag v1.0.0. ACTION Sheet not read. Provisional envelope used instead, flagged provisional=True.
critical F7 — Compiled data failed source audit - 10 of 31 rows [resolved] EVIDENCE (a) 3 rows labelled Sr are Eu-152+154. (b) 4 Niu 2022 rows claim Langmuir Qmax from a paper tabulating none. (c) Na-MK misattributed from another paper. (d) K-MK is a verbatim copy of K-MKBFS - a sorbent that does not exist. ACTION Quarantined/relabelled/re-typed as named constants in code.

medium F8 — Unmined master sheets add nothing [resolved]

EVIDENCE

Adsorption_Isotherms and Cs_Isotherms duplicate Functional_Data with the same fabricated rows; only 3 new values, which are cobalt.

ACTION

Marked duplicate. Not ingested.

high F9 — Surface area does not predict CATION uptake [open]

EVIDENCE

Cation-only, per source - lin2026 -0.90 (n=4), xiang2021 -0.90 (n=5), oulu2026 -0.47 (n=21), tarnovsky2024 +0.22 (n=6, inconclusive). Pooled cations +0.08. Varon: BET +0.19 vs Al-IV +0.95.

ACTION

Al-IV is the candidate primary feature. NOTE - an earlier claim of "six datasets confirm a negative BET correlation" was an OVERSTATEMENT, corrected here.

high F10 — Pooling adsorbates flips the sign of the BET relationship [resolved]

EVIDENCE

cations corr(BET,capacity) = +0.08; dyes = +0.66.

ACTION

Never pool adsorbate classes. Dyes use surface area; cations use charge sites. pooling_warning() enforces the check.

high F11 — Sr_Immobilization_Mechanisms is untraceable and largely duplicated [resolved]

EVIDENCE

Audit 2026-07-20 - the ONLY master sheet with no Source column at all, plus 66.7% internal duplication among its numeric cells. Same signature as the fabricated Na-MK/K-MK rows in F7.

ACTION

Relabelled FALSE / DISCARD in the warehouse. Not usable.

medium F12 — The v2 "conflict" was precision, not disagreement [resolved]

EVIDENCE

All 22 differing cells are Comp_Strength in rows 843-866 and differ only by rounding (34.36 vs 34.35777). N data.xls genuinely reports up to 6 dp, and 18 of the 24 v2 values appear verbatim upstream. So v2 preserved source precision and Physical_Data rounded to 2 dp.

ACTION

Q1 closed. v2 status suspect -> verified. Physical_Data remains canonical; the difference (≤ 0.005 MPa) is immaterial for modelling.

medium F13 — Several context sheets cite papers we do not hold [open]

EVIDENCE

Audit 2026-07-20 - Kim_2026, Yildirim_2024 and Duque_Redondo_2023 are cited by Surface_Complexation_Sr, MD_Simulation_Sr_Cs, Zeolite_Comparison_Sr_Cs, Sr_Leaching_Data, EXAFS and XRD sheets. None are in papers/references. Surface_Complexation cites "Eq. (10)-(12)" - equations, not sources, so those rows are model output rather than measurement. FTIR is 88.9% duplicated, XRD 72.7%.

ACTION

All remain UNVERIFIED and excluded from modelling. Acquiring the three papers is the only way to promote them; recorded as Q6.

high F14 — A high R2 Langmuir fit can leave Qmax essentially unconstrained [open]

EVIDENCE

Baek 2018 chabazite: $Q_0L = 1249.5$ mg/g with $K_L = 7.9e-05$ L/mg and $R^2 = 0.99989$, while the paper's own kinetics report equilibrium uptake of just 1.184 mg/g. K_L is two orders of magnitude below the other two zeolites in the same table. At low coverage the Langmuir isotherm is near-linear, so it fits almost perfectly while the plateau is extrapolated $\sim 1000\times$ beyond any measured point.

ACTION

Ingested but flagged - it exceeds the schema bound of 1000 mg/g and validate() reports it. Must be excluded from any fit unless the extrapolation is defended. Generalises: R^2 is not evidence that a fitted Q_{max} is meaningful. Screen future Langmuir rows on K_L and on the measured concentration range, not R^2 alone.

critical F15 — Fitted capacities cannot be validated - C0 was never captured [resolved with open gap]

EVIDENCE

Applying the F14 screen to all 35 fitted-isotherm rows: 32 of the 33 rows with a usable K_L have NO recorded initial concentration, so Langmuir saturation ($\theta = K \cdot C_0 / (1 + K \cdot C_0)$) cannot be computed. The screen runs on exactly one row (GF600, $\theta = 0.93$). Chabazite was caught only because its paper stated the equilibrium uptake in prose.

ACTION

initial_conc_mg_L made required for fitted-isotherm rows going forward; existing rows flagged c0_unknown. PROGRESS 2026-07-20: C_0 recovered for Baek (2658 mg/L) and El-Naggar (1000 mg/L), taking screenable rows from 1 to 10 - of which 8 are sound, 1 borderline (stilbite, $\theta = 0.775$) and 1 a confirmed artifact (chabazite, $\theta = 0.174$). 25 rows still lack C_0 . COMPLETE 2026-07-20: C_0 recovered for 7 sources; 29 of 35 rows now screenable. 27 sound, 1 borderline (stilbite 0.775), 1 artifact (chabazite 0.174). Six rows remain unscreenable because their methods state no isotherm range. Lesson recorded THREE times: K_L alone is not a screen - saturation is $K \cdot C_0 / (1 + K \cdot C_0)$, so a small K is compensated by a large C_0 . The same instinct would have discarded 19 sound rows.

medium F16 — $\theta = 1 - R_L$, so the saturation screen needs no C0 [resolved]

EVIDENCE

The Langmuir separation factor $R_L = 1 / (1 + b \cdot C_0)$ is reported by many papers as a favourability check. Our saturation measure $\theta = b \cdot C_0 / (1 + b \cdot C_0)$ is exactly $1 - R_L$. Verified against the independent C_0 route on six El-Naggar rows: the two derivations agree to three decimals. Resolved Tian ESPR ($\theta = 0.944$ Cs, 0.994 Sr) with back-calculated C_0 5.25 and 9.77 mmol/L, matching the Fig. 5 axes.

ACTION

Use R_L as the PRIMARY screen and C_0 recovery only as fallback. Found by rendering PDF figures and reading them visually - a capability previously and wrongly described as "manual figure reading" and deferred to Mulham.

high **F17 — Zhang 2021 rows are extrapolations with an unphysical intercept** [resolved]

EVIDENCE

Figure 8 read visually. The slope reproduces the published Qmax exactly ($1/11.196 = 89.32$; $1/22.464 = 44.52$), but both intercepts are NEGATIVE (-0.5365, -2.0942), which in $Ce/qe = 1/(Q_{max} \cdot b) + Ce/Q_{max}$ implies $b < 0$ and is unphysical; the table nonetheless reports positive b. Data reach only $Ce = 3.45$ and 3.75 mg/L , giving $\theta = 0.365$ and 0.074 - the slag fit reaches 7% of its plateau, worse than the chabazite artefact at 17%.

ACTION

Both rows flagged as extrapolations and excluded from modelling alongside chabazite. Screen now COMPLETE at 35 of 35: 31 sound, 1 borderline, 3 extrapolated.

high **F18 — The Oulu 29Si deconvolution was already published; it explains our failed fit** [resolved]

EVIDENCE

Table 2 of Hossain 2026 (doi:10.1016/j.matdes.2026.115471, page 6) reports the deconvolved Si environments for all 12 selected samples - the values our own fit attempted to derive. Found by rendering the page and reading it. The authors use TEN environments (Q0, Q1, Q2(1Al), Q2(0Al), Q4(4Al), Q3(1Al), Q4(3Al), Q3/Q4(2Al), Q3/Q4(1Al), Q4(0Al)) including Q3 and Q2 species with non-bridging oxygens. Our model used FIVE Q4(mAl) Gaussians only, which cannot represent that intensity; the optimiser pushed it outward and drove Q4(3Al) to zero.

ACTION

Confirms the NMR rejection was correct AND for the right reason - the chemical sanity check detected a structurally wrong peak model that $R^2 = 0.998$ and both pre-registered gates could not. Pipeline closed. The Oulu letter is no longer needed for this purpose. Values transcribed into src/geomind/data/nmr_ari.py; 11 of 12 rows sum to exactly 100%, one to 99.0 (published as such).

high **F19 — The Al-richness descriptor is valid only within a structural class, and the Oulu backfill does not widen its evidence base** [resolved]

EVIDENCE

Backfilling ARI from F18 attached the descriptor to 39 pool rows / 12 sorbents. Split by structure, the 12 sorbents behave as two different materials. Ca-FREE (n=4, Q4 45-73%): ARI +0.932, framework-only ARI +0.940 - uptake spans 2.13-11.53 mg/g (sd 3.98). Ca-BEARING (n=8, Q4 0-26%, Q0-Q2 dominated): ARI -0.108, Si/Al -0.135, Ca/Al -0.104, Q4% -0.247, BET +0.462 - NOTHING predicts, and uptake is nearly flat at $7.27 \pm 1.31 \text{ mg/g}$ regardless of composition. Pooling all 12 gives ARI +0.536, a mediocre value that conceals two regimes. Ca acts as a network modifier: there is no condensed aluminosilicate framework for a framework descriptor to measure, and uptake proceeds by a different mechanism (C-A-S-H exchange), roughly composition-independent.

ACTION

Restrict the ARI claim to Q4-rich framework aluminosilicates and state the structural gate explicitly. Generalises F9 one level down: never pool ADSORBATE classes, and never pool SORBENT structural classes either. IMPORTANT correction to my own estimate - I projected this backfill would widen the descriptor to ~19 sorbents; in fact only 4 of the 12 lie in the regime where it applies, so the evidence base is unchanged at Varon n=7 (Sr) + Oulu n=4 (NH4+). Al^{IV} scarcity remains the binding constraint on M4/M5.

critical **F20 — NMR characterisation and adsorption-capacity measurement are near-disjoint in this literature - Al^{IV} scarcity is structural, not a sampling accident** [open]

EVIDENCE

Corpus-wide classification of all 30 PDFs by keyword balance (adsorption terms vs leaching/immobilisation terms) against NMR content. Of the 6 papers carrying substantial solid-state NMR, FOUR are immobilisation/wasteform studies (Nevin 2026 nmr=54, Kim 2026 nmr=66, Vettese 2025 nmr=33, Kurumisawa 2021 nmr=13), ONE is adsorption (Hossain/Oulu nmr=11, but NH4+ and dyes only - no Cs/Sr), and ONE is mixed (Varon 2025 nmr=32). Meanwhile ALL 13 papers classified as adsorption studies report ZERO solid-state NMR. Varon is the single bridge between the two - which is exactly why Al^{IV} occurs in precisely 7 pool rows, all from Varon. These are two different research communities: water-treatment groups measure isotherms and BET and never run MAS NMR; nuclear-wasteform groups run full 29Si/27Al/133Cs NMR and measure leach rates and diffusivities, never isotherms.

ACTION

More scraping CANNOT fix Al^{IV} scarcity - the co-occurrence is rare by the structure of the field, not by chance, so Q7 has no answer within the adsorption literature. Two routes remain: (a) measure NMR ourselves on sorbents of known capacity, or (b) REFRAME the M4/M5 target property from adsorption capacity to leach resistance, where NMR is abundant. Note that route (b) reverses earlier triage - Kurumisawa 2021 and J. Environ. Manage. 370 (2024) were both REJECTED as "diffusivity/leaching studies", i.e. we discarded precisely the papers richest in the descriptor we need. The project aim ("immobilisation of Cs-137/Sr-90") is arguably better served by leach resistance than by adsorption capacity anyway; the two have been conflated. Decision required from PI - see reports/q7-descriptor-hunt-report.md.

high **F21 — Sr_Leaching_Data audited against source for the first time - LI verified, De corrupted to zero, and the Yildirim rows are misattributed** [resolved]

EVIDENCE

Both cited sources (Kim 2026, Nevin 2026) are now known to be on disk (see F13), so this sheet became auditable. Line-by-line against Kim Table 2 (p10) and Nevin p6: (a) LEACHABILITY INDEX - VERIFIED, 6 of 6 determinable rows exact. Kim NaSr1 15.5, NaSr2 14.8, NaSr3 16.0, KSr3 16.0; Nevin Sr_1 18.7, Sr_3 19.1. KSr1/KSr2 are correctly recorded as below detection - Kim states Sr leachate was under the ICP-OES limit (0.001 ppm), yielding non-detectable De and LI. (b) De_cm2_s - CORRUPTED. Every row reads 0.0, whereas Kim Table 2 reports 8.88E-16, 3.80E-14, 2.10E-16, 6.36E-16, 5.80E-18 and so on. Precision was lost to zero. Kim's De is also TIME-RESOLVED over five points (2 h, 7 h, 7 d, 14 d, 21 d), so no single scalar De pairs with the averaged LI in the first place. (c) CN_Sr_EXAFS on the two Yildirim_2024 rows - MISATTRIBUTED. The sheet gives Sr_Pre_1 = 7.87 and Sr_Post_1 = 7.95 sourced to Yildirim_2024. Kim p26 states the NaSr3 coordination number "was 7.87 +/- 0.69 before the leaching test" and "7.95 +/- 0.43 afterwards" - these are Kim's own NaSr3 pre/post values, and Kim attributes the pre-leach figure to reference [7], which is Geddes et al., J. Hazard. Mater. 488 (2025) 137426 - NOT Yildirim. The KSr3 value 7.55 is correctly attributed.

ACTION

Promote the LI column to VERIFIED_TRUE - it is the first immobilisation target variable in the project to survive source audit. Quarantine De_cm2_s entirely (re-extract from Kim Table 2 as a time series if needed). Quarantine both Yildirim_2024 rows as misattributed - third instance of the F7 pattern.

high **F22 — Pool B built and audited - LI verified twice over, but it has almost no compositional variation paired with NMR** **[resolved]**

EVIDENCE

Pool B v1 = 16 rows / 2 sources (Kim 2026, Nevin 2026), of which 8 carry a retention target, 2 are left-censored, and 6 are structure-only. (a) The LI = -log10(De) relation is now confirmed against TWO independent published tables - Kim Table 2 (12 of 13 cells to within 0.06) and Nevin Table 2 (4 of 4: 18.66/19.11/13.27/13.22 vs published 18.7/19.1/13.3/13.2). (b) A PUBLISHED ERRATUM was found. Kim Table 2 cell NaSr1 @ 2 h prints "8.88 E-16 (15.5)" but -log10(8.88e-16) = 15.05. Kim's own column Average LI of 15.5 equals mean(15.05, 15.94) exactly; had the cell truly been 15.5 the average would read 15.7. The printed (15.5) is a typo for (15.1). Verified at 430 dpi, so it is not a rendering artefact. The column average is unaffected. (c) The master sheet Sr_Leaching_Data carried only Nevin's two Sr rows and SILENTLY DROPPED both Cs rows (Cs_1 LI 13.3, Cs_3 LI 13.2) - a 50% loss of that table, undetectable without reading the source. (d) Nevin's headline result is preserved: in the SAME K-A-S-H gel, Sr is retained roughly six orders of magnitude better than Cs (De 7.8e-20 vs 6.0e-14). Nuclide identity dominates structure entirely at fixed composition.

ACTION

LIMITATION, stated plainly: only 4 rows pair an NMR descriptor with a retention target, and across those 4 the ARI is nearly constant (2.738-2.855) because Nevin is ONE formulation at Si/Al = 3. Kim varies activator (Na vs K) and loading but reports no Q4 deconvolution. So Pool B is currently as descriptor-starved as Pool A, for the opposite reason - Pool A has composition without NMR, Pool B has NMR without composition. Neither supports a descriptor model yet. This raises the priority of Q8 (Geddes 2025) further: it is the only identified source that may carry composition, NMR and a retention/binding measure together.

medium **F23 — Not every published deconvolution assigns m uniquely - Vettese's ARI is indeterminate and must not be computed** **[resolved]**

EVIDENCE

Vettese 2025 Table 4 (p7) reports a 29Si deconvolution, but its first two columns carry AMBIGUOUS assignments: "Q3(1Al) or Q4(4Al)" (m = 1 or 4) and "Q3 or Q4(3Al)" (m = 0 or 3). Computing the Al-richness index therefore has two permitted answers for the unleached sample: ARI = 2.289 if the Q4 readings are taken, or 0.861 if the Q3 readings are - a factor of 2.66. Populations do sum correctly (100.0 / 100.1 / 100.1), so the table is internally sound; it is the ASSIGNMENT that is undetermined, not the fit. Hossain and Nevin both assign m uniquely; Vettese does not.

ACTION

Do NOT compute or store an ARI for Vettese. A pipeline that derived ARI from any Q^n(mAl) table without checking assignment uniqueness would silently inject a 2.7x error into the one descriptor the project depends on. Vettese can still contribute structure-only rows (raw populations) and its Cs leach data, which is figure-only and single-formulation. Any future adapter must carry an assignment-ambiguity flag.

critical **F24 — Kurumisawa 2021 recovered - a rejected paper that bridges both pools - and it exposes a 12-order-of-magnitude naming collision in "diffusion coefficient"** **[resolved]**

EVIDENCE

Kurumisawa 2021 (doi:10.1617/s11527-021-01758-y) was REJECTED during batch triage as "a diffusivity study (diffus 65)". Reading it inline shows that verdict was wrong on content: (a) Figure 3b publishes the Langmuir MONOLAYER Cs CAPACITY for five metakaolin geopolymers made with five different alkali activators (K11 0.315, K9 0.425, 0.66K 0.530, N11 0.727, KN11 0.393 mmol/g) - a Pool A capacity WITH real compositional variation, which Pool A is short of. Pool A: 121 -> 126 rows. (b) Figure 2 publishes a Cs diffusivity for the same five specimens - Pool B. Pool B: 16 -> 26 rows. (c) THE COLLISION. Kurumisawa's is an INGRESS coefficient (Cs diffusing INTO an immersed specimen), reported in m2/s and of order 4e-12 m2/s = 4e-8 cm2/s. Kim's and Nevin's are ANSI/ANS 16.1 EFFECTIVE diffusivities (nuclide leaching OUT of a doped wasteform), of order 1e-16 and 1e-19 cm2/s. Same name, opposite experiment, TWELVE orders of magnitude apart. (d) Table 3 is the 29Si NMR of the ACTIVATOR SOLUTION (Q0-Q3 of the silicate), NOT of the hardened gel - so this paper supplies no framework Al^IV despite being NMR-bearing. (e) Applying the F14/F16 saturation screen to the recovered capacities: K11 theta 0.962 sound, KN11 0.847 sound, K9 0.704 borderline, 0.66K 0.608 borderline, and N11 theta 0.499 an EXTRAPOLATION ARTEFACT - N11 being the HIGHEST reported capacity of the five.

ACTION

Both halves ingested through separate adapters under D13. RetentionType gains INGRESS_DIFFUSIVITY as a distinct member and pooling_warning() now raises an explicit CATEGORY ERROR if both diffusivity kinds appear together. N11 flagged as an artefact. All Kurumisawa rows carry from_figure=True. LESSON: the batch triage rejected this paper on keyword balance without reading it, and keyword balance cannot see a capacity that lives only in a figure. The other triage rejections should be re-read the same way - starting with J. Environ. Manage. 370 (2024).

medium **F25 — The last triage rejection (J. Environ. Manage. 370) is upheld as data, but it is a high-value BIBLIOGRAPHY that names the paired-data sources both pools lack** **[resolved]**

EVIDENCE

D15 re-read of Mukiza et al., J. Environ. Manage. 370 (2024) 122746, "Recent advances in immobilization of Cs/Sr-bearing wastes in AAMs: a review". It is a REVIEW, not primary data: zero Langmuir/Qmax/Kd of its own, and its Table 2 explicitly states "the data are not directly comparable". So the batch rejection was CORRECT for direct ingestion - none of its tabulated numbers may enter either pool as primary rows. BUT reading it surfaced exactly the paired-descriptor sources F20/F22 said were rare: (a) Komljenovic et al. 2020, J. Hazard. Mater. 388, doi:10.1016/j.jhazmat.2019.121765, "Immobilization of cesium with alkali-activated blast furnace slag" - the review states it ties 29Si NMR speciation (Q1, Q2(0Al), Q2(1Al) sites, Al^IV/Si of C-A-S-H) to strength AND Cs. NMR + composition + Cs in one paper. Same group (Provis) as Nevin/Geddes. A second candidate F20 bridge alongside Geddes 2025. (b) Qian et al. 2001, J. Nucl. Mater. 299, 199-204, doi:10.1016/S0022-3115(01)00700-0 - Cs and Sr distribution ratios R_d (3.9e3->6.4e3 mol/g Cs; 6.4e3->1.2e4 mol/g Sr). A K_d measure = Pool A. (c) Jang et al. 2016, J. Hazard. Mater. 318, 339-346, doi:10.1016/j.jhazmat.2016.07.003, "Physical barrier effect ... on diffusivity of Cs and Sr" - a Cs/Sr diffusivity (Pool B) source, and it is ALREADY ON DISK (papers/references/Physical barrier effect ...pdf), registered but never mined.

ACTION

Keep the review out of both pools as a data source. Use it as a source index. Q9 opened for Komljenovic 2020 (acquire), Q10 for Qian 2001 (acquire), and Jang 2016 queued for Pool B extraction since it is already held. This is the third time (after F24 Kurumisawa and the Kim/Nevin content-index miss) that real value was found by reading a document already in hand rather than acquiring a new one.

medium **F26 — Jang 2016 mined into Pool B - 14 rows, both nuclides, with pore structure; and it confirms the LI aggregation trap a third time** **[resolved]**

EVIDENCE

Jang, Park & Lee, J. Hazard. Mater. 318 (2016) 339-346, doi:10.1016/j.jhazmat.2016.07.003 - the Q11 source, already on disk, now extracted. Seven doped waste forms (F1/F2/F3 fly-ash geopolymers, S1/S3/S5 fly-ash+slag blends, PC Portland baseline), each with Cs AND Sr leachability indices (Table 7) and pore structure (Table 5). Pool B: 26 -> 40 rows / 4 sources. (a) LI AGGREGATION CONFIRMED a third time. Jang's LI = mean over intervals of -log10(De_n): F1 Cs interval De's give 10.02 ~ published 10.0, whereas -log10(mean De) = 9.46 would be wrong by 0.5. Vindicates schema trap 4. (b) Sr LI > Cs LI in EVERY one of the 7 specimens - the universal Cs<Sr retention result, now across 7 more matrices, consistent with Nevin's ~6-order gap. (c) Si/Al deliberately NOT stored. The precursor-bulk Si/Al is computable (1.55-2.13) but omits the sodium-silicate activator Si, so it is not the gel Si/Al, and the F/S series use different silicate loadings (0.25 vs 0.17) so it is not even comparable across specimens. Storing it would repeat the F1 transplant error in a new form. Recorded as provenance-only constants. (d) Three new pore-structure columns added (porosity, capillary pore volume, critical pore diameter); Jang links these to De. corr(porosity, Cs LI) = -0.27 across the 6 geopolymers - weak, consistent with Jang's own point that critical pore DIAMETER matters more than total porosity.

ACTION

Ingested via _from_jang2016 as 14 LI rows. Mean De values retained as documented adapter constants (uncorrupted, unlike Kim's) but not emitted as separate rows, to avoid implying an independent measurement - a diffusivity target can regenerate them later. PC baseline flagged matrix_class=portland_cement, trivially filterable. Q11 closed. Fourth on-disk recovery of the session.

critical **F27 — Varon's leaching companion paper provides a WITHIN-SAMPLE causal test - framework Al^{IV} concentration governs Sr K_D, and destroying it by leaching destroys sorption** **[resolved]**

EVIDENCE

Batch-4 paper Varon et al., Open Ceramics 25 (2026) 100895, doi:10.1016/j.oceram.2025.100895 - the leaching companion to varon2025 (Cleaner Materials). Its 0 h data are IDENTICAL to varon2025 (verified: K-0.5-16 BET 72, AlIV 96.0%, KD 4100) and are NOT re-ingested; the 96 h water-leached state is new. Tables 3-4 give, for all 7 samples at 0 h and 96 h: BET, [AlIV]_exp (mmol/g), AlIV/V/V_I %, full Q4(mAl) speciation, Sr KD and Qmax. RESULT: [AlIV] CONCENTRATION (mmol/g) is the strongest predictor of Sr KD: corr = +0.946 (n=7), rising to +0.986 excluding the barely-reacted outlier K-1.2-8 (H₂O/M₂O=8, BET 15, Qmax 0.02). This beats AlIV% (+0.80), ARI (+0.77) and crushes BET (+0.19, or -0.24 without the outlier) and Si/Al (-0.84). THE CAUSAL TEST: leaching removes framework Al, and corr(Δ[AlIV], ΔKD) = +0.938 - the samples that lose the most framework Al on leaching lose the most sorption. K-0.5-16 loses 1.39 mmol/g AlIV and 2750 mL/g KD (largest of both); K-1.5-16 loses 0.04 mmol/g AlIV and the least KD. This is the same sample losing capacity AS it loses exchange sites - close to a causal demonstration that tetrahedral Al^{IV} is the Sr exchange site, not merely a correlate.

ACTION

Ingested 7 NEW 96h-leached Sr KD rows via _from_varon_leached2026 (Pool A 126 -> 133 rows / 12 sources), sorbent_name "... (leached 96h)" so they can never be pooled with the fresh varon2025 rows. The "Interplay..." PDF in the same batch is a confirmed DUPLICATE of varon2025 (skipped). Strengthens the project's central finding: framework Al predicts cation uptake, and this is the first WITHIN-SAMPLE (not just cross-sample) evidence for it. Note [AlIV] CONCENTRATION > AlIV% > ARI as the ranking - absolute exchange-site density beats the per-Si fraction.

medium **F28 — Stanojevic 2025 - 2 Cs leachability rows for a fly-ash geopolymer; Cs passes the LI criterion yet >70% leaches, reaffirming Cs mobility** **[resolved]**

EVIDENCE

Batch-4. Stanojevic et al., Constr. Build. Mater. 500 (2025) 144217, doi:10.1016/j.conbuildmat.2025.144217 (Komljenovic/Provis group). Fly-ash geopolymer, Cs doped at 2% and 5%, semi-dynamic leach. Lmean (Fig. 6b) = 10.3 (2%Cs) and 9.9 (5%Cs) over 90 days (9.7/9.3 over the first 5 days) - both clear the ANSI/ANS-16.1 / NRC floor of 6. BUT cumulative Cs leached exceeds 70% after 5 days (71.0% / 76.5%): the LI criterion is passed while most of the Cs still leaves - Cs remains highly mobile. 29Si shows Q4(mAl) (fly-ash framework), and the higher 5% Cs loading introduces less-crosslinked Q1,2 units (Cs slows condensation), a qualitative echo of the framework story.

ACTION

Ingested 2 Pool B rows via _from_stanojevic2025 (40 -> 42 rows / 5 sources), from_figure=True (Lmean read from the Fig. 6b annotation). One fly-ash geopolymer at two Cs LOADINGS (a control variable), so no composition variation; the Q4 data is figure-only and not digitised into per-sample descriptor rows. Reinforces the universal Cs<Sr pattern: even a passing LI hides >70% Cs release.

medium **F29 — Komljenovic 2020 (Q9) closed - it is a Ca-system (C-A-S-H) and QUANTITATIVELY confirms F19 - the aluminosilicate framework immobilises Cs better than a Ca gel** **[resolved]**

EVIDENCE

Batch-4, Q9 acquisition. Komljenovic et al., J. Hazard. Mater. 388 (2020) 121765, doi:10.1016/j.jhazmat.2019.121765. Alkali-activated blast-furnace slag (AABFS), Cs doped 2%/5%. The hoped-for "framework NMR + Cs" bridge it is NOT: AABFS is a C-A-S-H binder whose 29Si NMR shows only low-connectivity Qn(mAl) sites (n=1-3, m=0-1) with NO framework Q4(mAl) - exactly the Ca-modifier regime of F19, so the descriptor is C-A-S-H chain length / AlIV/Si, not ARI. Cs preferentially associates with the N-(C)-A-S-H gel over the C-A-S-H gel. QUANTITATIVE F19 CONFIRMATION: Cs Lmean (Fig. 5b) = 7.8 (2%) and 7.0 (5%), both above the ANSI/ANS-16.1 floor of 6, but markedly LOWER than the framework geopolymers - fly-ash (Stanojevic 10.3/9.9; Jang ~10) and metakaolin (Nevin Cs 13.2/13.3). The calcium-silicate gel holds Cs less well than the aluminosilicate framework.

ACTION

Q9 CLOSED. Ingested 2 Pool B Cs LI rows via _from_komljenovic2020 (42 -> 44 rows / 6 sources), matrix_class slag_blend, from_figure. Note for the eventual paper: Cs leachability index now stratifies cleanly by binder chemistry - framework MK ~13 > framework FA ~10 > Ca-rich slag ~7.5 - the immobilisation-side echo of the adsorption-side framework-Al finding (F27). Komljenovic is the Provis group; same lineage as Stanojevic, Nevin, Geddes.

medium **F30 — Qian 2001 (Q10) closed - Sr/Cs distribution ratios from a scanned 2001 paper, read visually; supports the framework-Al story from the Ca side** **[resolved]**

EVIDENCE

Batch-4, Q10 acquisition. Qian, Sun & Tay, J. Nucl. Mater. 299 (2001) 199-204, doi:10.1016/S0022-3115(01)00700-0. SCANNED PDF (no text layer) - values read visually from the page-4 body text. Aluminium-rich alkali-activated slag (AAS) and M-AAS (AAS + metakaolin + zeolite + NaOH-treated attapulgite clay), a Ca-rich C-A-S-H binder. Distribution ratio Rd (0.002 M CsCl/SrCl₂, 7 d): AAS Cs 3.9e3, Sr 6.4e3; M-AAS Cs 6.4e3, Sr 1.2e4. Adding Al-rich metakaolin/clay RAISES Rd for both ions (M-AAS > AAS by 67-88%), and both beat OPC by 1.5-3x - i.e. more aluminosilicate exchange capacity = more uptake, even in a calcium-silicate binder. This is the framework-Al mechanism seen from the Ca side. UNIT FLAG: the paper labels Rd "mol/g", but Rd is a distribution coefficient whose magnitude (3.9e3-1.2e4) is consistent with mL/g. Stored in kd_mL_g with the caveat recorded, not silently "corrected".

ACTION

Q10 CLOSED. Ingested 4 Pool A KD rows (AAS/M-AAS x Cs/Sr) via _from_qian2001 (133 -> 137 rows / 13 sources), sorbent_class AAM (slag C-A-S-H, not framework). Fifth on-disk/visual recovery of the batch; second successful visual read of a scanned/figure-locked source this session.

medium **F31 — Arbel-Haddad 2022 mined - 6 Cs cumulative-leached rows for MK framework geopolymers; DOI unverifiable from the AAM and flagged as such** **[resolved]**

EVIDENCE

Batch-4. Arbel-Haddad, Harnik, Schlosser & Goldbourt, "Cesium immobilization in metakaolin-based geopolymers elucidated by ¹³³Cs solid state NMR spectroscopy" (J. Nucl. Mater., ~2022). Metakaolin geopolymer, Si/Al = 1 (very Al-rich framework), Na activator with Cs substituting Na at 2.5-100%. Table 4 gives the cumulative fraction of Cs leached (48 h, two-step): 2.5% 0.044, 5% 0.008, 7.5% 0.004, 25% 0.006, 50% 0.056, 100% 0.139. Leaching is LOWEST (best immobilisation) at 5-25% Cs and rises steeply at high loading - a non-monotonic loading effect. The ¹³³Cs NMR resolves Cs SITES (structural water / framework-bound), not a Si framework descriptor, so this yields a target but no ARI. DATA-INTEGRITY NOTE: no DOI is present in this author-accepted manuscript (every DOI in the file is a bibliography reference). Provenance is recorded as a flagged citation string, NOT a guessed DOI, per the standing no-fabrication rule.

ACTION

Ingested 6 Pool B rows via _from_arbelhaddad2022 (44 -> 50 rows / 7 sources), introducing the CUMULATIVE_LEACHED_FRACTION retention type (lower = better). Provenance flagged "(DOI-unverified-AAM)"; confirm and correct the DOI when the published version is available. Loading recorded as null (Cs% is a molar Na substitution, not a wt%).

low **F32 — O'Donoghue 2026 assessed but NOT ingested - a purely structural study with no sorption or retention target on any sample** **[resolved]**

EVIDENCE

Batch-4. O'Donoghue, Geddes, Wilkinson, Stennett, Kim, Iuga, Hayes & Walkley, "High-field multinuclear MAS NMR and synchrotron XANES reveal the influence of strontium salt chemistry on geopolymer nanostructure", Dalton Trans. 55 (2026) 10673, doi:10.1039/d6dt00775a (the Sheffield group; same lineage as Nevin/Geddes). It varies the Sr SALT (Sr(OH)2.8H2O, SrCO3, Sr(NO3)2, SrSO4) and Sr LOADING (Sr/Al 0.025-0.125) at FIXED Si/Al = 1.5, and reports full Q4(mAl) deconvolutions (SI Tables S1-S4, 3 d and 28 d) plus XANES. But there is NO quantitative Sr target anywhere - no capacity, K_d, leachability index or diffusivity (searched: all zero). Every row would be descriptor-only.

ACTION

NOT INGESTED, by the same discipline as D7 (do not pad a pool with rows that have no usable target). Unlike Nevin's structure-only rows - which sit alongside companion rows that DO carry an LI - none of O'Donoghue's samples have a target, and Si/Al is fixed so even the descriptor does not vary across the design. It remains a useful structural reference (how Sr salt chemistry and SrCO3 formation perturb the K-A-S-H framework) and is catalogued in papers/MANIFEST.md. If a Sr uptake/retention measurement on these exact samples ever appears, revisit.

medium **F33 — Foams 2020 (Petlitzkaia) extracted at last - and the bare framework foam out-capacities the ferrocyanide-functionalised one in pure water** **[resolved]**

EVIDENCE

Petlitzkaia et al., J. Cleaner Production 269 (2020) 122400, doi:10.1016/j.jclepro.2020.122400 (CEA/Poulesquen). The batch-3 deferred item (mangled text layer), now read visually. Two metakaolin K-geopolymer foams: GF (bare) and FGF (functionalised with K2Cu(Fe(CN)6), a Prussian-blue analogue). Cs capacity is matrix-dependent, so THREE distinct quantities appear and are kept apart: isotherm MAX in pure water (GF 250, FGF 175 mg/g); single-point uptake at 100 ppm (GF 80, FGF 65 - a kinetics endpoint, NOT ingested); fresh-water capacity with competitive K/Na/Mg/Ca (GF 150, FGF 100). The isotherm reaches and passes saturation (capacity declines beyond Cs_eq 3.92e-3 mol/L), so the F14/theta screen is satisfied - these are genuine maxima, not extrapolations. MECHANISM SPLIT worth noting: GF removes Cs by FRAMEWORK ion exchange (Na<->Cs); FGF adds ferrocyanide (K<->Cs). In pure water the BARE FRAMEWORK out-capacities the functionalised foam (250 > 175 mg/g); FGF wins only on SELECTIVITY (Kd ~3.5e5 vs ~3.5e4 mL/g). Consistent with the framework-Al story - the aluminosilicate exchange sites carry the raw capacity; functionalisation trades some for selectivity.

ACTION

Ingested 4 Pool A rows via _from_petlitzkaia2020 (137 -> 141 rows / 14 sources): 2 pure-water maxima + 2 fresh-water (competing_ion flagged). Single-point 80/65 not ingested (redundant kinetics endpoint). from_figure. Batch-3 Foams deferral CLOSED. FGF is a functionalised sorbent - flag before pooling its capacity with bare framework geopolymers.

critical **F34 — Geddes 2025 (Q8) acquired - the ATOMIC-LEVEL MECHANISM behind the framework-Al finding; Sr binds at extra-framework Al^IV charge-balancing sites** **[resolved]**

EVIDENCE

The critical Q8 acquisition, delivered by the supervisor after the Checkpoint-2 request. Geddes, Stennett, Wilkinson, Griffith, Hanna, Provis & Walkley, "Alkali-mediated Sr incorporation mechanism and binding capacity of alkali aluminosilicate hydrate in geopolymers", J. Hazard. Mater. 488 (2025) 137426, doi:10.1016/j.jhazmat.2025.137426 (Sheffield/White Rose). Metakaolin N-A-S-H geopolymer, Si/Al = 1.5, Na (and Na/K) activator, Sr(OH)2.8H2O doped 0-3.5 wt%. Multi-technique: ²⁹Si/²⁷Al/²³Na MAS NMR + Sr K-edge XANES/EXAFS. MECHANISM (the key result): at low loading Sr2+ is incorporated into EXTRA-FRAMEWORK CHARGE-BALANCING SITES bonded via oxygen to tetrahedral Al^IV (SiIV-O-AlIV), with a brewsterite-Sr-like local structure - i.e. Sr sits exactly at the Al^IV sites the framework-Al descriptor counts. The N-A-S-H gel is Q4(4Al)/Q4(3Al)-dominant (Al-rich framework, Table 6). BINDING CAPACITY: Sr is chemically bound up to ~0.5-1.5 wt% Sr(OH)2.8H2O (~2-5 mg Sr per g); above the upper bound the excess precipitates as SrCO3 / Sr(OH)2. A doped-in chemical-binding limit set by the framework charge, NOT a sorption-from-solution capacity or a leachability index.

ACTION

Q8 CLOSED - the highest-value acquisition of the project, and it delivers exactly the mechanism F20/F27 needed. NOT ingested into either pool: the binding capacity is a distinct doped-in quantity (framework-charge-limited chemical incorporation), and Si/Al is fixed so there is no composition-varying descriptor-vs-target series (same no-shoehorning discipline as O'Donoghue F32). Its value is as the ATOMIC MECHANISM: it upgrades the central finding from "correlation + within-sample causal test (F27)" to "...+ direct atomic-scale proof that Sr binds at the Al^IV charge-balancing sites the descriptor measures". Cite as the mechanistic capstone of the descriptor-and-methodology paper.

medium **F35 — Frederickx 2025 mined - 4 Cs/Sr leachability rows for a real-waste N-A-S-H form; and the accelerated NH4NO3 test is shawn to flatten the Cs/Sr difference** **[resolved]**

EVIDENCE

Batch-5. Frederickx, Mukiza, Phung et al., Sustainability 17 (2025) 1756, doi:10.3390/su17041756 (SCK CEN). A real Cs/Sr-rich nitrate waste immobilised in a metakaolin N-A-S-H alkali-activated matrix at two waste loadings. Table 5 gives accelerated (NH4NO3) leachability indices: Cs 6.9/7.0, Sr 7.0/6.7, all above the ACRIA floor of 6. METHODOLOGICAL POINT worth citing: in the accelerated NH4NO3 test Cs and Sr LIs are the same scale (~7), but the paper's own deionised-water extrapolation (Fig 5, one month) gives Cs 9.6 and Sr 12.5 - Sr >> Cs, the usual pattern. The NH4NO3 medium was built for Ca-cement and over-attacks divalent Sr, so the accelerated test UNDERSTATES Sr immobilisation and hides the Cs/Sr gap.

ACTION

Ingested 4 Pool B rows via _from_frederickx2025 (50 -> 54 rows / 8 sources) with leachant="NH4NO3_accelerated" so they are never pooled with the deionised-water LIs of the other sources (same leachant-confound discipline as the ingress/effective diffusivity split, F24). The deionised-water equivalents (Cs 9.6, Sr 12.5) are derived/extrapolated and NOT ingested, only noted. ALSO: the third file in this batch (CEMCON-D-25-01825, doi:10.1016/j.cemconres.2026.108214) is the clean manuscript of Kim 2026 - a DUPLICATE of the already-ingested kim2026 source; skipped.

Appendix B

Pool A — every adsorption row

All 141 rows across 14 sources. ‘—’ = not reported for that row. Generated directly from the pipeline; nothing hand-entered.

#	source	sorbent	adsorbate	class	cap (mg/g)	type	Si/Al	BET	Al ^{IV} %	provenance
1	literature_co mpilation	Na-MKBFS	Cs	radionuc lide	75.8	langmuir_ qmax	2.13	—	—	El-Naggar_2018
2	literature_co mpilation	Na-MKBFS	Cs	radionuc lide	80.5	langmuir_ qmax	2.13	—	—	El-Naggar_2018
3	literature_co mpilation	Na-MKBFS	Cs	radionuc lide	82.2	langmuir_ qmax	2.13	—	—	El-Naggar_2018
4	literature_co mpilation	K-MKBFS	Cs	radionuc lide	42.2	langmuir_ qmax	2.4	—	—	El-Naggar_2018
5	literature_co mpilation	K-MKBFS	Cs	radionuc lide	48	langmuir_ qmax	2.4	—	—	El-Naggar_2018
6	literature_co mpilation	K-MKBFS	Cs	radionuc lide	61.5	langmuir_ qmax	2.4	—	—	El-Naggar_2018
7	literature_co mpilation	MU-PMK80	Cs	radionuc lide	59.6	langmuir_ qmax	1.43	—	—	Hamed_2025
8	literature_co mpilation	MU-PMK80	Eu	other	18.6	langmuir_ qmax	1.43	—	—	Hamed_2025
9	literature_co mpilation	MU-PMK80	Cs	radionuc lide	59.4	langmuir_ qmax	1.43	—	—	Hamed_2025
10	literature_co mpilation	MU-PMK80	Eu	other	20	langmuir_ qmax	1.43	—	—	Hamed_2025
11	literature_co mpilation	MU-PMK80	Cs	radionuc lide	57.7	langmuir_ qmax	1.43	—	—	Hamed_2025
12	literature_co mpilation	MU-PMK80	Eu	other	21.6	langmuir_ qmax	1.43	—	—	Hamed_2025
13	literature_co mpilation	MU-PMK80	Cs	radionuc lide	59.6	langmuir_ qmax	1.43	—	—	Hamed_2025
14	literature_co mpilation	MU-PMK80	Cs	radionuc lide	59.4	langmuir_ qmax	1.43	—	—	Hamed_2025
15	literature_co mpilation	MU-PMK80	Cs	radionuc lide	57.7	langmuir_ qmax	1.43	—	—	Hamed_2025
16	literature_co mpilation	MU-PMK80	Cs	radionuc lide	59.6	langmuir_ qmax	1.43	—	—	Hamed_2025
17	literature_co mpilation	FA-Geopolymer	Cs	radionuc lide	112	langmuir_ qmax	2	—	—	Tian_2019
18	literature_co mpilation	FA-Geopolymer	Sr	radionuc lide	24.2	langmuir_ qmax	2	—	—	Tian_2019
19	literature_co mpilation	Zeolite-Geopolymer	Cs	radionuc lide	87.9	langmuir_ qmax	2	—	—	Zheng_2023
20	literature_co mpilation	M/SZMs	Cs	radionuc lide	104	freundlich_ qmax	2	—	—	Lei_2021
21	literature_co mpilation	M/SZMs	Sr	radionuc lide	54.9	langmuir_ qmax	2	—	—	Lei_2021
22	oulu2026	Si20 Al1 Na1	NH4+	analogu e	2.13	single_poi nt_q	20	249	—	doi:10.1016/j.matd e...
23	oulu2026	Si15 Al1 Na1	NH4+	analogu e	4.27	single_poi nt_q	15	242	—	doi:10.1016/j.matd e...
24	oulu2026	Si10 Al1 Na1	NH4+	analogu e	5.47	single_poi nt_q	10	206	—	doi:10.1016/j.matd e...
25	oulu2026	Si5Al1Na1	NH4+	analogu e	7	single_poi nt_q	5	132	—	doi:10.1016/j.matd e...
26	oulu2026	Si4Al1Na1	NH4+	analogu e	8.07	single_poi nt_q	4	109	—	doi:10.1016/j.matd e...
27	oulu2026	Si3Al1Na1	NH4+	analogu e	9.13	single_poi nt_q	3	71.6	—	doi:10.1016/j.matd e...
28	oulu2026	Si2Al1Na1	NH4+	analogu e	11.5	single_poi nt_q	2	31	—	doi:10.1016/j.matd e...
29	oulu2026	Si1Al1Na1	NH4+	analogu e	13.7	single_poi nt_q	1	8.27	—	doi:10.1016/j.matd e...
30	oulu2026	Si20 Al1 Na1 Ca21	NH4+	analogu e	7.73	single_poi nt_q	20	6.37	—	doi:10.1016/j.matd e...
31	oulu2026	Si15 Al1 Na1 Ca16	NH4+	analogu e	6.87	single_poi nt_q	15	6.04	—	doi:10.1016/j.matd e...
32	oulu2026	Si10 Al1 Na1 Ca11	NH4+	analogu e	5.33	single_poi nt_q	10	5.14	—	doi:10.1016/j.matd e...
33	oulu2026	Si5 Al1 Na1 Ca6	NH4+	analogu e	4.53	single_poi nt_q	5	3.52	—	doi:10.1016/j.matd e...
34	oulu2026	Si4 Al1 Na1 Ca5	NH4+	analogu e	5.93	single_poi nt_q	4	6.92	—	doi:10.1016/j.matd e...
35	oulu2026	Si3 Al1 Na1 Ca4	NH4+	analogu e	6.67	single_poi nt_q	3	10.3	—	doi:10.1016/j.matd e...
36	oulu2026	Si2 Al1 Na1 Ca3	NH4+	analogu e	8.87	single_poi nt_q	2	13.2	—	doi:10.1016/j.matd e...
37	oulu2026	Si1 Al1 Na1 Ca2	NH4+	analogu e	11	single_poi nt_q	1	14.1	—	doi:10.1016/j.matd e...
38	oulu2026	Si4 Al1 Na1 Ca1	NH4+	analogu e	6.53	single_poi nt_q	4	2.79	—	doi:10.1016/j.matd e...
39	oulu2026	Si4 Al1 Na1 Ca2	NH4+	analogu e	6.8	single_poi nt_q	4	1.03	—	doi:10.1016/j.matd e...
40	oulu2026	Si4 Al1 Na1 Ca3	NH4+	analogu e	8.27	single_poi nt_q	4	2.7	—	doi:10.1016/j.matd e...
41	oulu2026	Si4 Al1 Na1 Ca4	NH4+	analogu e	8.67	single_poi nt_q	4	7.95	—	doi:10.1016/j.matd e...
42	oulu2026	Si4 Al1 Na1 Ca5	NH4+	analogu e	5.93	single_poi nt_q	4	6.92	—	doi:10.1016/j.matd e...
43	oulu2026	Si20 Al1 Na1	methylene_bl ue	dye	90.3	single_poi nt_q	20	249	—	doi:10.1016/j.matd e...
44	oulu2026	Si15 Al1 Na1	methylene_bl ue	dye	87.6	single_poi nt_q	15	242	—	doi:10.1016/j.matd e...
45	oulu2026	Si10 Al1 Na1	methylene_bl ue	dye	77.2	single_poi nt_q	10	206	—	doi:10.1016/j.matd e...
46	oulu2026	Si5Al1Na1	methylene_bl ue	dye	56.6	single_poi nt_q	5	132	—	doi:10.1016/j.matd e...

#	source	sorbent	adsorbate	class	cap (mg/g)	type	Si/Al	BET	Al ^{IV} %	provenance
47	oulu2026	Si4Al1Na1	methylene_blue	dye	48.4	single_point_q	4	109	—	doi:10.1016/j.matd
48	oulu2026	Si3Al1Na1	methylene_blue	dye	36.7	single_point_q	3	71.6	—	doi:10.1016/j.matd
49	oulu2026	Si2Al1Na1	methylene_blue	dye	14.3	single_point_q	2	31	—	doi:10.1016/j.matd
50	oulu2026	Si1Al1Na1	methylene_blue	dye	3.57	single_point_q	1	8.27	—	doi:10.1016/j.matd
51	oulu2026	Si20 Al1 Na1 Ca21	methylene_blue	dye	27	single_point_q	20	6.37	—	doi:10.1016/j.matd
52	oulu2026	Si15 Al1 Na1 Ca16	methylene_blue	dye	25.2	single_point_q	15	6.04	—	doi:10.1016/j.matd
53	oulu2026	Si10 Al1 Na1 Ca11	methylene_blue	dye	21.6	single_point_q	10	5.14	—	doi:10.1016/j.matd
54	oulu2026	Si5 Al1 Na1 Ca6	methylene_blue	dye	14.2	single_point_q	5	3.52	—	doi:10.1016/j.matd
55	oulu2026	Si4 Al1 Na1 Ca5	methylene_blue	dye	20.3	single_point_q	4	6.92	—	doi:10.1016/j.matd
56	oulu2026	Si3 Al1 Na1 Ca4	methylene_blue	dye	26.8	single_point_q	3	10.3	—	doi:10.1016/j.matd
57	oulu2026	Si2 Al1 Na1 Ca3	methylene_blue	dye	31.4	single_point_q	2	13.2	—	doi:10.1016/j.matd
58	oulu2026	Si1 Al1 Na1 Ca2	methylene_blue	dye	34.9	single_point_q	1	14.1	—	doi:10.1016/j.matd
59	oulu2026	Si4 Al1 Na1 Ca1	methylene_blue	dye	11.2	single_point_q	4	2.79	—	doi:10.1016/j.matd
60	oulu2026	Si4 Al1 Na1 Ca2	methylene_blue	dye	9.3	single_point_q	4	1.03	—	doi:10.1016/j.matd
61	oulu2026	Si4 Al1 Na1 Ca3	methylene_blue	dye	11.5	single_point_q	4	2.7	—	doi:10.1016/j.matd
62	oulu2026	Si4 Al1 Na1 Ca4	methylene_blue	dye	27.2	single_point_q	4	7.95	—	doi:10.1016/j.matd
63	oulu2026	Si4 Al1 Na1 Ca5	methylene_blue	dye	20.3	single_point_q	4	6.92	—	doi:10.1016/j.matd
64	oulu2026	Si20 Al1 Na1	rhodamine_6G	dye	53.7	single_point_q	20	249	—	doi:10.1016/j.matd
65	oulu2026	Si15 Al1 Na1	rhodamine_6G	dye	28	single_point_q	15	242	—	doi:10.1016/j.matd
66	oulu2026	Si10 Al1 Na1	rhodamine_6G	dye	13.5	single_point_q	10	206	—	doi:10.1016/j.matd
67	oulu2026	Si5Al1Na1	rhodamine_6G	dye	7.74	single_point_q	5	132	—	doi:10.1016/j.matd
68	oulu2026	Si4Al1Na1	rhodamine_6G	dye	6.29	single_point_q	4	109	—	doi:10.1016/j.matd
69	oulu2026	Si3Al1Na1	rhodamine_6G	dye	5.93	single_point_q	3	71.6	—	doi:10.1016/j.matd
70	oulu2026	Si2Al1Na1	rhodamine_6G	dye	5.21	single_point_q	2	31	—	doi:10.1016/j.matd
71	oulu2026	Si1Al1Na1	rhodamine_6G	dye	2.31	single_point_q	1	8.27	—	doi:10.1016/j.matd
72	oulu2026	Si20 Al1 Na1 Ca21	rhodamine_6G	dye	36.3	single_point_q	20	6.37	—	doi:10.1016/j.matd
73	oulu2026	Si15 Al1 Na1 Ca16	rhodamine_6G	dye	33.4	single_point_q	15	6.04	—	doi:10.1016/j.matd
74	oulu2026	Si10 Al1 Na1 Ca11	rhodamine_6G	dye	27.6	single_point_q	10	5.14	—	doi:10.1016/j.matd
75	oulu2026	Si5 Al1 Na1 Ca6	rhodamine_6G	dye	12.4	single_point_q	5	3.52	—	doi:10.1016/j.matd
76	oulu2026	Si4 Al1 Na1 Ca5	rhodamine_6G	dye	14.6	single_point_q	4	6.92	—	doi:10.1016/j.matd
77	oulu2026	Si3 Al1 Na1 Ca4	rhodamine_6G	dye	17.1	single_point_q	3	10.3	—	doi:10.1016/j.matd
78	oulu2026	Si2 Al1 Na1 Ca3	rhodamine_6G	dye	18.9	single_point_q	2	13.2	—	doi:10.1016/j.matd
79	oulu2026	Si1 Al1 Na1 Ca2	rhodamine_6G	dye	20.4	single_point_q	1	14.1	—	doi:10.1016/j.matd
80	oulu2026	Si4 Al1 Na1 Ca1	rhodamine_6G	dye	9.18	single_point_q	4	2.79	—	doi:10.1016/j.matd
81	oulu2026	Si4 Al1 Na1 Ca2	rhodamine_6G	dye	7.74	single_point_q	4	1.03	—	doi:10.1016/j.matd
82	oulu2026	Si4 Al1 Na1 Ca3	rhodamine_6G	dye	10.3	single_point_q	4	2.7	—	doi:10.1016/j.matd
83	oulu2026	Si4 Al1 Na1 Ca4	rhodamine_6G	dye	19.3	single_point_q	4	7.95	—	doi:10.1016/j.matd
84	oulu2026	Si4 Al1 Na1 Ca5	rhodamine_6G	dye	14.6	single_point_q	4	6.92	—	doi:10.1016/j.matd
85	katada2024	FAU 6.1	Cs	radionuclide	73.1	iec_meq_g	1.37	—	—	10.1021/acs.langmui...
86	katada2024	LTA 6.7	Cs	radionuclide	23.9	iec_meq_g	1.12	—	—	10.1021/acs.langmui...
87	katada2024	MFI 1.2	Cs	radionuclide	133	iec_meq_g	12.3	—	—	10.1021/acs.langmui...
88	katada2024	YFI 1.3	Cs	radionuclide	110	iec_meq_g	11.1	—	—	10.1021/acs.langmui...
89	katada2024	MOR 1.5	Cs	radionuclide	191	iec_meq_g	9.88	—	—	10.1021/acs.langmui...
90	katada2024	MOR 1.7	Cs	radionuclide	211	iec_meq_g	8.37	—	—	10.1021/acs.langmui...
914	tarnovsky2021	Kaolin	Cs	radionuclide	13.3	langmuir_qmax	1.03	9	—	10.15407/hftp15.01....
924	tarnovsky2022	Kaolin	Sr	radionuclide	14.9	langmuir_qmax	1.03	9	—	10.15407/hftp15.01....
934	tarnovsky2023	GP 1	Cs	radionuclide	233	langmuir_qmax	1.95	26	—	10.15407/hftp15.01....
944	tarnovsky2024	GP 1	Sr	radionuclide	48.2	langmuir_qmax	1.95	26	—	10.15407/hftp15.01....
954	tarnovsky2025	GP 2	Cs	radionuclide	124	langmuir_qmax	1.59	88	—	10.15407/hftp15.01....

#	source	sorbent	adsorbate	class	cap (mg/g)	type	Si/Al	BET	Al ^{IV} %	provenance
964	tarnovsky2024	GP 2	Sr	radionuclide	57	langmuir_qmax	1.59	88	—	10.15407/hftp15.01...
97	varon2025	K-0.5-16	Sr	radionuclide	—	kd_mL_g	1.14	72	96	10.1016/j.clema.202...
98	varon2025	K-1-16	Sr	radionuclide	—	kd_mL_g	1.55	163	93.8	10.1016/j.clema.202...
99	varon2025	K-1.2-16	Sr	radionuclide	—	kd_mL_g	1.67	188	94	10.1016/j.clema.202...
1000	varon2025	K-1.5-16	Sr	radionuclide	—	kd_mL_g	1.79	212	91.4	10.1016/j.clema.202...
1001	varon2025	K-1.2-8 (inferred)	Sr	radionuclide	—	kd_mL_g	1.83	15	88	10.1016/j.clema.202...
1002	varon2025	K-1.2-12 (inferred)	Sr	radionuclide	—	kd_mL_g	1.63	182	92.4	10.1016/j.clema.202...
1003	varon2025	Na-1.2-16 (inferred)	Sr	radionuclide	—	kd_mL_g	1.67	38	84.7	10.1016/j.clema.202...
104	lin2026	ACSr	Sr	radionuclide	222	iec_meq_g	—	—	—	10.3390/ceramics902...
105	lin2026	NCGO	Cs	radionuclide	54.5	iec_meq_g	—	39.2	—	10.3390/ceramics902...
106	lin2026	NiGO	Cs	radionuclide	46.5	iec_meq_g	—	38.8	—	10.3390/ceramics902...
107	lin2026	CoGO	Cs	radionuclide	45.2	iec_meq_g	—	29.3	—	10.3390/ceramics902...
108	lin2026	Ba-4A	Sr	radionuclide	167	iec_meq_g	—	—	—	10.3390/ceramics902...
109	lin2026	CuFC/4A	Cs	radionuclide	86.4	iec_meq_g	—	5.25	—	10.3390/ceramics902...
110r	tian2019_espr	Zeolite A (from fly ash)	Cs	radionuclide	282	langmuir_qmax	0.821	121	—	10.1007/s11356-019-...
111r	tian2019_espr	Zeolite A (from fly ash)	Sr	radionuclide	169	langmuir_qmax	0.821	121	—	10.1007/s11356-019-...
112	xiang2021	GF300	Cs	radionuclide	162	single_poi nt_q	1.02	53.8	—	10.1016/j.jece.2021...
113	xiang2021	GF400	Cs	radionuclide	164	single_poi nt_q	1.02	49.4	—	10.1016/j.jece.2021...
114	xiang2021	GF500	Cs	radionuclide	186	single_poi nt_q	1.02	45.8	—	10.1016/j.jece.2021...
115	xiang2021	GF600	Cs	radionuclide	192	single_poi nt_q	1.02	37.8	—	10.1016/j.jece.2021...
116	xiang2021	GF600	Cs	radionuclide	216	langmuir_qmax	1.02	37.8	—	10.1016/j.jece.2021...
117	zhang2021	Flyash-geomaterial	Cs	radionuclide	89.3	langmuir_qmax	—	—	—	10.3390/app11188407
118	zhang2021	Slag-geomaterial	Cs	radionuclide	44.5	langmuir_qmax	—	—	—	10.3390/app11188407
119	baek2018	chabazite	Cs	radionuclide	1.25e+03	langmuir_qmax	1.5	—	—	10.1016/j.micromeso...
120	baek2018	stilbite	Cs	radionuclide	76.3	langmuir_qmax	3.88	—	—	10.1016/j.micromeso...
121	baek2018	heulandite	Cs	radionuclide	59.5	langmuir_qmax	3.48	—	—	10.1016/j.micromeso...
122	kurumisawa2021	K11	Cs	radionuclide	41.9	langmuir_qmax	—	—	—	10.1617/s11527-021-...
123	kurumisawa2021	K9	Cs	radionuclide	56.5	langmuir_qmax	—	—	—	10.1617/s11527-021-...
124	kurumisawa2021	0.66K	Cs	radionuclide	70.4	langmuir_qmax	—	—	—	10.1617/s11527-021-...
125	kurumisawa2021	N11	Cs	radionuclide	96.6	langmuir_qmax	—	—	—	10.1617/s11527-021-...
126	kurumisawa2021	KN11	Cs	radionuclide	52.2	langmuir_qmax	—	—	—	10.1617/s11527-021-...
127	varon_leached2026	K-0.5-16 (leached 96h)	Sr	radionuclide	51.7	experimental_max	1.27	109	82.4	10.1016/j.oceram.20...
128	varon_leached2026	K-1-16 (leached 96h)	Sr	radionuclide	35	experimental_max	1.52	181	92.1	10.1016/j.oceram.20...
129	varon_leached2026	K-1.2-16 (leached 96h)	Sr	radionuclide	35.9	experimental_max	1.63	199	91.9	10.1016/j.oceram.20...
130	varon_leached2026	K-1.5-16 (leached 96h)	Sr	radionuclide	21	experimental_max	1.77	221	93.6	10.1016/j.oceram.20...
131	varon_leached2026	K-1.2-8 (leached 96h)	Sr	radionuclide	44.7	experimental_max	1.63	28	88.8	10.1016/j.oceram.20...
132	varon_leached2026	K-1.2-12 (leached 96h)	Sr	radionuclide	48.2	experimental_max	1.63	220	90.4	10.1016/j.oceram.20...
133	varon_leached2026	Na-1.2-16 (leached 96h)	Sr	radionuclide	24.5	experimental_max	1.63	44	85.8	10.1016/j.oceram.20...
134	qian2001	AAS	Cs	radionuclide	—	kd_mL_g	—	—	—	10.1016/S0022-3115(...

#	source	sorbent	adsorbate	class	cap (mg/g)	type	Si/Al	BET	Al ^{IV} %	provenance
1 3 5	qian2001	AAS	Sr	radionuclide	—	kd_mL_g	—	—	—	10.1016/S0022-3115(...
1 3 6	qian2001	M-AAS	Cs	radionuclide	—	kd_mL_g	—	—	—	10.1016/S0022-3115(...
1 3 7	qian2001	M-AAS	Sr	radionuclide	—	kd_mL_g	—	—	—	10.1016/S0022-3115(...
1 3 8	petlitzkaia2020	GF (geopolymer foam, pure water)	Cs	radionuclide	250	experimental_max	—	—	—	10.1016/j.jclepro.2...
1 3 9	petlitzkaia2020	FGF (functionalised geopolymer foam, pure water)	Cs	radionuclide	175	experimental_max	—	—	—	10.1016/j.jclepro.2...
1 4 0	petlitzkaia2020	GF (geopolymer foam, fresh water)	Cs	radionuclide	150	experimental_max	—	—	—	10.1016/j.jclepro.2...
1 4 1	petlitzkaia2020	FGF (functionalised geopolymer foam, fresh water)	Cs	radionuclide	100	experimental_max	—	—	—	10.1016/j.jclepro.2...

Appendix C

Pool B — every immobilisation row

All 54 rows across 8 sources. ‘target’ distinguishes a leachability index (higher = better) from an ingress diffusivity; the two are never pooled. ‘cens.’ marks a left-censored (below-detection, best-performing) result.

#	source	matrix	class	nuclide	target	value	cens.	ARI	poros.%	provenance
1	kim2026	NaSr1	na_geopolymer	Sr	leachability index	15.5		—	—	10.1016/j.cemconres...
2	kim2026	NaSr2	na_geopolymer	Sr	leachability index	14.8		—	—	10.1016/j.cemconres...
3	kim2026	NaSr3	na_geopolymer	Sr	leachability index	16		—	—	10.1016/j.cemconres...
4	kim2026	KSr1	k_geopolymer	Sr	leachability index	—	left	—	—	10.1016/j.cemconres...
5	kim2026	KSr2	k_geopolymer	Sr	leachability index	—	left	—	—	10.1016/j.cemconres...
6	kim2026	KSr3	k_geopolymer	Sr	leachability index	16		—	—	10.1016/j.cemconres...
7	nevin2026	C_0	k_geopolymer	none	—	—		2.81	—	10.1039/d5ta90211h
8	nevin2026	Sr_1	k_geopolymer	Sr	leachability index	18.7		2.75	—	10.1039/d5ta90211h
9	nevin2026	Sr_3	k_geopolymer	Sr	leachability index	19.1		2.8	—	10.1039/d5ta90211h
10	nevin2026	Cs_1	k_geopolymer	Cs	leachability index	13.3		2.74	—	10.1039/d5ta90211h
11	nevin2026	Cs_3	k_geopolymer	Cs	leachability index	13.2		2.85	—	10.1039/d5ta90211h
12	nevin2026	C_0	k_geopolymer	none	—	—		2.81	—	10.1039/d5ta90211h
13	nevin2026	Sr_1	k_geopolymer	Sr	—	—		2.93	—	10.1039/d5ta90211h
14	nevin2026	Sr_3	k_geopolymer	Sr	—	—		2.81	—	10.1039/d5ta90211h
15	nevin2026	Cs_1	k_geopolymer	Cs	—	—		2.8	—	10.1039/d5ta90211h
16	nevin2026	Cs_3	k_geopolymer	Cs	—	—		2.76	—	10.1039/d5ta90211h
17	kurumisawa2021	K11	k_geopolymer	Cs	d ingress cm2 s	4.70e-08		—	—	10.1617/s11527-021-...
18	kurumisawa2021	K11	k_geopolymer	Cs	d ingress cm2 s	4.00e-08		—	—	10.1617/s11527-021-...
19	kurumisawa2021	K9	k_geopolymer	Cs	d ingress cm2 s	7.00e-09		—	—	10.1617/s11527-021-...
20	kurumisawa2021	K9	k_geopolymer	Cs	d ingress cm2 s	4.80e-09		—	—	10.1617/s11527-021-...
21	kurumisawa2021	0.66K	k_geopolymer	Cs	d ingress cm2 s	4.40e-08		—	—	10.1617/s11527-021-...
22	kurumisawa2021	0.66K	k_geopolymer	Cs	d ingress cm2 s	4.00e-08		—	—	10.1617/s11527-021-...
23	kurumisawa2021	N11	na_geopolymer	Cs	d ingress cm2 s	8.30e-09		—	—	10.1617/s11527-021-...
24	kurumisawa2021	N11	na_geopolymer	Cs	d ingress cm2 s	4.70e-09		—	—	10.1617/s11527-021-...
25	kurumisawa2021	KN11	k_geopolymer	Cs	d ingress cm2 s	1.90e-08		—	—	10.1617/s11527-021-...
26	kurumisawa2021	KN11	k_geopolymer	Cs	d ingress cm2 s	2.00e-08		—	—	10.1617/s11527-021-...
27	jang2016	FA1 geopolymer	fly_ash_geopolymer	Cs	leachability index	10		—	29.9	10.1016/j.jhazmat.2...
28	jang2016	FA1 geopolymer	fly_ash_geopolymer	Sr	leachability index	14.1		—	29.9	10.1016/j.jhazmat.2...
29	jang2016	FA2 geopolymer	fly_ash_geopolymer	Cs	leachability index	10.1		—	33.9	10.1016/j.jhazmat.2...
30	jang2016	FA2 geopolymer	fly_ash_geopolymer	Sr	leachability index	14.1		—	33.9	10.1016/j.jhazmat.2...
31	jang2016	FA3 geopolymer	fly_ash_geopolymer	Cs	leachability index	10.2		—	31.9	10.1016/j.jhazmat.2...
32	jang2016	FA3 geopolymer	fly_ash_geopolymer	Sr	leachability index	14.4		—	31.9	10.1016/j.jhazmat.2...
33	jang2016	FA3+BFS 90:10	slag_blend	Cs	leachability index	7.1		—	39.7	10.1016/j.jhazmat.2...
34	jang2016	FA3+BFS 90:10	slag_blend	Sr	leachability index	11.5		—	39.7	10.1016/j.jhazmat.2...

#	source	matrix	class	nuclide	target	value	cens.	ARI	poros.%	provenance
3 5	jang2016	FA3+BFS 70:30	slag_blend	Cs	leachability index	8		—	31.9	10.1016/j.jhazmat.2016.05.011
3 6	jang2016	FA3+BFS 70:30	slag_blend	Sr	leachability index	12.7		—	31.9	10.1016/j.jhazmat.2016.05.011
3 7	jang2016	FA3+BFS 50:50	slag_blend	Cs	leachability index	8		—	25.9	10.1016/j.jhazmat.2016.05.011
3 8	jang2016	FA3+BFS 50:50	slag_blend	Sr	leachability index	12.1		—	25.9	10.1016/j.jhazmat.2016.05.011
3 9	jang2016	Portland cement	portland_cement	Cs	leachability index	7		—	28.3	10.1016/j.jhazmat.2016.05.011
4 0	jang2016	Portland cement	portland_cement	Sr	leachability index	9.8		—	28.3	10.1016/j.jhazmat.2016.05.011
4 1 5	stanojevic2021	G-FA-2%Cs	fly_ash_geopolymer	Cs	leachability index	10.3		—	—	10.1016/j.conbuildmat.2021.126441
4 2 5	stanojevic2021	G-FA-5%Cs	fly_ash_geopolymer	Cs	leachability index	9.9		—	—	10.1016/j.conbuildmat.2021.126441
4 3 020	komljenovic2020	AABFS-2%Cs	slag_blend	Cs	leachability index	7.8		—	—	10.1016/j.jhazmat.2020.105202
4 4 020	komljenovic2020	AABFS-5%Cs	slag_blend	Cs	leachability index	7		—	—	10.1016/j.jhazmat.2020.105202
4 5 022	arbelhaddad2022	MK-GP 2.5%Cs	na_geopolymer	Cs	clf	0.044		—	—	Arbel-Haddad 2022 J. Hazard. Mater.
4 6 022	arbelhaddad2022	MK-GP 5%Cs	na_geopolymer	Cs	clf	0.008		—	—	Arbel-Haddad 2022 J. Hazard. Mater.
4 7 022	arbelhaddad2022	MK-GP 7.5%Cs	na_geopolymer	Cs	clf	0.004		—	—	Arbel-Haddad 2022 J. Hazard. Mater.
4 8 022	arbelhaddad2022	MK-GP 25%Cs	na_geopolymer	Cs	clf	0.006		—	—	Arbel-Haddad 2022 J. Hazard. Mater.
4 9 022	arbelhaddad2022	MK-GP 50%Cs	na_geopolymer	Cs	clf	0.056		—	—	Arbel-Haddad 2022 J. Hazard. Mater.
5 0 022	arbelhaddad2022	MK-GP 100%Cs	na_geopolymer	Cs	clf	0.139		—	—	Arbel-Haddad 2022 J. Hazard. Mater.
5 1 5	frederickx2021	MK N-A-S-H waste form (low loading)	na_geopolymer	Cs	leachability index	6.9		—	—	10.3390/su17041756
5 2 5	frederickx2021	MK N-A-S-H waste form (low loading)	na_geopolymer	Sr	leachability index	7		—	—	10.3390/su17041756
5 3 5	frederickx2021	MK N-A-S-H waste form (high loading)	na_geopolymer	Cs	leachability index	7		—	—	10.3390/su17041756
5 4 5	frederickx2021	MK N-A-S-H waste form (high loading)	na_geopolymer	Sr	leachability index	6.7		—	—	10.3390/su17041756

Appendix D

Decisions register

The 16 standing project decisions, each with its reason. These are the rules the pipeline enforces.

#	Decision	Reason
D 1	Drop Si_AI and Density from physical data	F1 - transplanted, not measured
D 2	Re-scope M3b to cross-system transfer	Different material system; no viscosity; 92 mixes
D 3	Repo private, all-rights-reserved	Unpublished research
D 4	Copyrighted PDFs excluded from git	Copyright + 80 MB; manifest tracks them
D 5	Do not use Feasibility_Ranges	F6 - anti-correlated with ground truth
D 6	Kd kept in its own column, never in capacity_mg_g	Different physical quantity
D 7	Reject Tian supplement's 24-row table	No composition; would pad without features
D 8	Quarantine Na-MK, K-MK and all Niu 2022 rows	F7 - not in stated sources
D 9	No subagents for extraction or analysis	Missing project context lets silent errors in
D 10	Pre-register NMR acceptance at r>=0.9	Deconvolution is under-determined; stops retroactive tuning
D 11	ARI claims are gated on Q4-rich framework sorbents	F19 - Ca-bearing gels have no framework; descriptor is meaningless there, not merely weak
D 12	Never pool sorbent structural classes in a correlation	F19 - pooling hid ARI +0.932 and -0.108 behind a single +0.536
D 13	Run adsorption capacity and leach resistance as TWO parallel pools, never merged	PI decision 2026-07-20 on F20. Removal-from-solution and immobilisation are different applications with different targets; the descriptor is available only in the second. Consistent with D12.
D 15	Re-read every triage-rejected paper inline before trusting the rejection	F24 - Kurumisawa was rejected on keyword balance, but its Cs capacity lives only in a figure, which keyword counts cannot see
D 16	Never compute ARI from a deconvolution whose m-assignment is ambiguous	F23 - Vettese's Table 4 admits ARI 2.289 or 0.861, a 2.7x fork
D 14	Index the corpus by CONTENT, not filename	F13/F20 - Kim 2026 and Nevin 2026, the two most descriptor-rich papers held, were invisible to a filename-based check because their filenames are not in Latin script

Appendix E

Glossary of chemical terms

Term	Meaning
Alkali-activated material (AAM)	Binder made by activating an aluminosilicate precursor with a strong alkali; geopolymers are the aluminium-rich, low-calcium end of this family.
Al ^{IV} / framework aluminium	Aluminium in tetrahedral coordination within the network; each site carries a negative charge balanced by an exchangeable cation. The predictive descriptor of this project.
ARI (Al-richness index)	Mean number of Al neighbours per Si, $\sum_m m \cdot I_m / 100$, from ²⁹ Si NMR speciation; an NMR-based measure of framework aluminium.
BET surface area	Specific surface area from N ₂ physisorption (m ² g ^{−1}). Predicts dye (physisorption) uptake but <i>not</i> cation (ion-exchange) uptake.
Distribution coefficient K _d	Ratio of sorbed to solution concentration (mL g ^{−1}); a measure of affinity, distinct from a capacity.
Effective diffusivity D _e	Rate constant for a nuclide leaching out of a waste form (cm ² s ^{−1}); lower is better.
Leachability index (LI, L)	−log ₁₀ (D _e); ANSI/ANS 16.1 ranking of immobilisation. Higher = better; ≥6 is the acceptance floor.
Langmuir Q _{max}	Fitted monolayer saturation capacity (mg g ^{−1}); meaningful only if the experiment approached saturation (the θ screen).
Q ⁿ (mAl)	A silicon environment with <i>n</i> bridging oxygens, <i>m</i> of them to aluminium; from ²⁹ Si MAS NMR.
R _L / θ	Langmuir separation factor; θ = 1 − R _L is the fraction of saturation reached, used to screen fitted capacities.
Si/Al ratio	Molar silicon-to-aluminium ratio; lower means more charge-balancing sites.
Network modifier (Ca)	Calcium disrupts the aluminosilicate framework toward low-connectivity C–A–S–H, where framework descriptors lose meaning.